

Free AI Coding Setup Roadmap

Ollama + Claude Code + Antigravity — 12GB RAM Edition

Your Setup

RAM: 12GB

Recommended Model: qwen3:8b

Cost: FREE

PHASE 1 — Install Ollama

Step 1 — Download & Install Ollama

Linux / macOS:

```
curl -fsSL https://ollama.com/install.sh | sh
```

Windows:

Go to <https://ollama.com> → click Download → run the .exe installer

Step 2 — Verify installation

```
ollama --version
```

You should see a version number like: *ollama version 0.x.x*

Step 3 — Pull the best model for 12GB RAM

```
ollama pull qwen3:8b
```

Download is ~5GB. This is the best coding model that fits in 12GB RAM.

Alternative (lighter, faster):

```
ollama pull deepseek-r1:8b
```

Step 4 — Test Ollama is working

```
ollama run qwen3:8b
```

Type a message. If you get a reply, Ollama is working. Press Ctrl+D to exit.

PHASE 2 — Install Claude Code

Step 5 — Install Node.js (required)

Linux / macOS:

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt-get install -y nodejs
```

Windows: Download from <https://nodejs.org> → install LTS version

Step 6 — Install Claude Code via npm

```
npm install -g @anthropic-ai/claude-code
```

Step 7 — Verify Claude Code installed

```
claude --version
```

PHASE 3 — Connect Claude Code to Ollama (Free!)

Step 8 — Set environment variables

Linux / macOS — add these to ~/.bashrc or ~/.zshrc:

```
export ANTHROPIC_AUTH_TOKEN=ollama
export ANTHROPIC_API_KEY=ollama
export ANTHROPIC_BASE_URL=http://localhost:11434
```

Windows PowerShell:

```
$env:ANTHROPIC_AUTH_TOKEN="ollama"
$env:ANTHROPIC_API_KEY="ollama"
$env:ANTHROPIC_BASE_URL="http://localhost:11434"
```

Step 9 — Apply the changes (Linux/macOS only)

```
source ~/.bashrc
```

Step 10 — Launch Claude Code with Ollama

```
claude --model qwen3:8b --allow-dangerously-skip-permissions
```

You now have a free AI coding agent running locally!

PHASE 4 — Install & Integrate Antigravity

Step 11 — Download Antigravity

Go to <https://antigravity.google> → click Download for your OS

Sign in with your personal Gmail account (not Workspace). Free quota is automatic.

Step 12 — Open a project in Antigravity

Launch Antigravity → Open Folder → select your project directory

Step 13 — Configure Antigravity to use your local Ollama model

In Antigravity settings, look for 'Model Provider' or 'API Endpoint'.

```
Set the base URL to: http://localhost:11434
Set model to: qwen3:8b
Leave API key blank or type: ollama
```

Step 14 — Use Claude Code inside your project

In your project terminal inside Antigravity, run:

```
claude --model qwen3:8b --allow-dangerously-skip-permissions
```

Claude Code will now use Ollama as its backend — 100% free, fully local.

Tip	Details
Model too slow?	Try deepseek-r1:8b — it's faster on lower RAM
Daily free limit hit?	Antigravity lets you switch models — use Gemini Flash as backup
Offline test	Disconnect internet, run a prompt — if it works, you're truly local
Context window	qwen3:8b supports 32k tokens — good enough for most projects

Total cost: \$0.00 • Everything runs on your machine • No API key needed